CLAIMS

What is claimed is:

1. A portable data storage apparatus, comprising:
    - a ruggedized housing designed for physical transport;
    - a non-volatile data storage medium contained within the housing;
    - a self-contained power source contained within the housing; and
    - at least one massively parallel optical interface integrated into the housing for data input and output.

2. The apparatus of claim 1, comprising a first massively parallel optical interface for data input and a second, separate massively parallel optical interface for data output.

3. The apparatus of claim 2, wherein the first interface comprises a two-dimensional array of photon sensors and the second interface comprises a two-dimensional array of photon emitters.

4. The apparatus of claim 1, wherein the apparatus is configured to transfer data via the optical interface at a rate exceeding 1 terabit per second.

5. The apparatus of claim 1, further comprising a protective, sealed cover for the optical interface.